Reagent Rental Model

Biochemistry Analyzer – Reagent Rental Pricing Formula

We need a general formula with parameters. Let:

- C_a= analyzer landed cost
- i= annual cost of capital (interest rate)
- n= contract years (3)
- M_{total}= total maintenance & service cost over n years
- For each reagent j: T_j= total tests per month for that reagent (given as 102 for all, but could vary)
- R_j= tests per pack for reagent j
- $P_{cogs,j}$= your cost per pack (COGS)
- Over n years, total months = $12n$. Total tests for reagent j = $T_j \times 12n$. Number of packs needed (rounded up) $Q_j = \lceil \frac{T_j \times 12n}{R_j} \rceil$
- Total COGS over n years = $\sum_j Q_j \times P_{cogs,j}$
- Total fixed cost to recover = $C_a \times (1 + i \times n) + M_{total}$
- Total revenue needed = Fixed cost + Total COGS
- Markup factor = Total revenue needed / Total COGS
- New selling price per pack for reagent j = $P_{cogs,j}$ × Markup factor

Alternatively, you could set a fixed dollar markup per pack: Markup per pack = $\frac{C_a(1+in)+M_{total}}{\sum_j Q_j}$. Then new price = $P_{cogs,j}$ + Markup per pack. That would be uniform absolute markup.

*You can replace i and M_{total} with your actual values. The formula below works with any numbers.

1. **Step-by-Step Formula**

Step 1 – Total tests per reagent over 3 years

$$\text{Tests}_{total} = T_{month} \times 12 \times n$$

NB: Here the total number of Test performed per Parameter is the same. That is 102 for each and all the parameters.

Step 2 – Number of packs needed for each reagent (rounded up)

$$Q_j = \left\lceil \frac{\text{Tests}_{total}}{R_j} \right\rceil$$

where R_j= tests per pack for reagent j (from table).

Step 3 – Total reagent COGS over 3 years

$$\text{COGS}_{total} = \sum_{j=1}^{15} \left(Q_j \times P_{\text{cogs},j} \right)$$

where $P_{\text{cogs},j}$= your cost per pack (given in table).

Step 4 – Total fixed cost to recover (analyzer + interest + maintenance)

$$\text{Fixed Cost} = C_a \times (1 + i \times n) + M_{total}$$

Step 5 – Total revenue needed from reagent sales

$$\text{Revenue}_{needed} = \text{Fixed Cost} + \text{COGS}_{total}$$

Step 6 – Markup factor (same percentage for all reagents)

$$\text{Markup} = \frac{\text{Revenue}_{needed}}{\text{COGS}_{total}}$$

Step 7 – New selling price per pack for reagent *j*

$$\boxed{P_{\text{sell},j} = P_{\text{cogs},j} \times \text{Markup}}$$

Excel-Ready Formula (Single Cell for Markup)
Input cells:
B2 = C_a(8000)
B3 = i(0.12)
B4 = n(3)
B5 = M_{total}(2400)
B6:B20 = $P_{\text{cogs},j}$ for each reagent
C6:C20 = tests per pack R_j
D6:D20 = monthly tests (102)
Then markup factor:
text
= (B2*(1+B3*B4) + B5 + SUMPRODUCT(CEILING(D6:D20*12*B4 / C6:C20, 1), B6:B20)) /
SUMPRODUCT(CEILING(D6:D20*12*B4 / C6:C20, 1), B6:B20)
Multiply each $P_{\text{cogs},j}$ by this factor to get new selling price.

6. Final Business Rule

$$\boxed{\text{Selling Price per Pack} = \text{COGS per Pack} \times \frac{C_a(1 + in) + M_{total} + \sum(\lceil \frac{12n \times T_{month}}{R_j} \rceil \times \text{COGS}_j)}{\sum(\lceil \frac{12n \times T_{month}}{R_j} \rceil \times \text{COGS}_j)}}$$

Use this formula with your actual monthly test volumes, maintenance costs, and cost of capital. Do **not** accept fixed low reagent prices – they must cover the analyzer placement.

Example 1 = SHEET 1

PBS Biochemistry Analyzer – Reagent Rental Pricing Formula

(Refer to the Excel Sheet Labelled Labelled 1)

3-Year Placement |15 Parameters | Fixed Monthly Test Volumes

Numerical Calculation (with assumed $i = 12\%$, $M_{total} = \$2,400$)

Must Known Determing Factors

Parameters	Symbol	Value
Analyzer landed cost	C_a	$8,000 (Variables)
Contract period	n	3 years (Variables)
Annual cost of capital (interest rate)	i	12% (assumed) * (Variables)
Total maintenance & service cost over 3 years	M_{total}	$2,400 (assumed 10% of $C_a$per year) *
Operating months	$12n$	36 months (Variables)
Tests per month per parameter (from customer data)	T_{month}	102 (same for all 15 parameters)

1. **Step-by-Step Formula**

Step 1 – Total tests per reagent over 3 years

$$\text{Tests}_{total} = T_{month}\ x\ 12\ x\ n$$
$$= 102\ x\ 12\ x\ 3$$
$$= 3{,}672 \text{ tests}$$

Step 2 – Number of packs needed for each reagent (rounded up)

$$Q_j = \left\lceil \frac{\text{Tests}_{total}}{R_j} \right\rceil$$

where R_j= tests per pack for reagent j (from table).

Total packs over 3 years: $\sum Q_j = 114$

Step 3 – Total reagent COGS over 3 years

Sum of $Q_j \times P_{\text{cogs},j}$ (using given pack prices as COGS):

Total reagent COGS over 3 years

$$\text{COGS}_{total} = \sum_{j=1}^{15} (Q_j \times P_{\text{cogs},j})$$

where $P_{\text{cogs},j}$= your cost per pack (given in table).

COGS_{total}= **$7,809.31**

Fixed cost to recover

Step 4 – Total fixed cost to recover (analyzer + interest + maintenance)

$\text{Fixed Cost} = C_a \; x \; (1 + i \; x \; n) + M_{total}$

$\text{Fixed Cost} = 8{,}000 \; x \; (1 + 0.12 \; x \; 3) + 2{,}400$

$\text{Fixed Cost} = 8{,}000 \times 1.36 + 2{,}400$

$\text{Fixed Cost} = 10{,}880 + 2{,}400$

$\textbf{Fixed Cost} = \$\mathbf{13,280}$

Total revenue needed & markup

Step 5 – Total revenue needed from reagent sales

$\text{Revenue}_{needed} = \text{Fixed Cost} + \text{COGS}_{total}$

$\text{Revenue}_{needed} = \$\mathbf{13,280} + \$\mathbf{7,809.31}$

$\text{Revenue}_{needed} = \$\mathbf{21,089.13}$

Step 6 – Markup factor (same percentage for all reagents)

$$\text{Markup} = \frac{\text{Revenue}_{needed}}{\text{COGS}_{total}}$$

$\text{Markup} = \$\mathbf{21,089.31}$ / 7,809.31 =2.7006

Markup = **2.7006**

New selling price per pack for reagent *j*

$\boxed{P_{\text{sell},j} = P_{\text{cogs},j} \times \text{Markup}}$

Multiply each COGS by 2.7006:

As seen in the excel sheet

Reagent	COGS/pack ($)	New Selling Price ($/pack)
ALB	23.10	**62.38**
TP	31.61	**85.39**
ALP	50.43	**136.19**
ALT	32.56	**87.94**
AST	35.89	**96.93**
TBIL	42.06	**113.59**

Reagent	COGS/pack ($)	New Selling Price ($/pack)
DBIL	64.93	**175.36**
GGT	64.01	**172.86**
CHOL	53.55	**144.62**
HDL-C	123.38	**333.26**
LDL-C	77.12	**208.27**
TG	106.47	**287.57**
CREA	178.94	**483.23**
UA	46.11	**124.53**
UREA	49.24	**133.00**

These prices ensure that over 3 years you recover:

- Analyzer cost ($8,000)
- 12% annual interest on locked capital ($2,880 total)
- Maintenance & service ($2,400)
- All reagent COGS ($7,809)
- **Zero additional profit** (break-even). If you want profit, multiply final prices by (1 + profit margin).

Important Notes for PBS

- **Shelf life (1 year)** – The customer's monthly volume (102 tests per parameter) uses about 38 packs per year (114/3). This is well within the 1-year stability. No waste.
- **Open system** – The same formula applies regardless of reagent brand; just replace $P_{\text{cogs},j}$ with your actual cost.
- **Maintenance & interest** – Replace assumed numbers with your real costs.
- **Rounding** – Always round up packs; otherwise you would not recover full pack costs from supplier.

Example 2 = SHEET 2

PBS Biochemistry Analyzer – Reagent Rental Pricing Formula

(Refer to the Excel Sheet Labelled 2)

3-Year Placement |15 Parameters | Fixed Monthly Test Volumes

Numerical Calculation (with assumed $i = 12\%$, $M_{total} = \$2,400$)

Must Known Determing Factors

Parameters	**Symbol**	**Value**
Analyzer landed cost	C_a	\$8,000 (Variables)
Contract period	n	3 years (Variables)
Annual cost of capital (interest rate)	i	12% (assumed) * (Variables)
Total maintenance & service cost over 3 years	M_{total}	\$2,400 (assumed 10% of $C_a$per year) *
Operating months	$12n$	36 months (Variables)
Tests per month per parameter (from customer data)	T_{month}	50 (same for all 15 parameters)

2. **Step-by-Step Formula**

Step 1 – Total tests per reagent over 3 years

$$\begin{aligned} \text{Tests}_{total} &= T_{month}\ x\ 12\ x\ n \\ &= 50\ x\ 12\ x\ 3 \\ &= 1{,}800 \text{ tests} \end{aligned}$$

Step 2 – Number of packs needed for each reagent (rounded up)

$$Q_j = \left\lceil \frac{\text{Tests}_{total}}{R_j} \right\rceil$$

where R_j= tests per pack for reagent j (from table).

Total packs over 3 years: $\sum Q_j = 58$

Step 3 – Total reagent COGS over 3 years

Sum of $Q_j \times P_{\text{cogs},j}$ (using given pack prices as COGS):

Total reagent COGS over 3 years

$$\text{COGS}_{total} = \sum_{j=1}^{15} (Q_j \times P_{\text{cogs},j})$$

where $P_{\text{cogs},j}$= your cost per pack (given in table).

COGS_{total}= **\$4021.55**

Fixed cost to recover

Step 4 – Total fixed cost to recover (analyzer + interest + maintenance)

Fixed Cost $= C_a\ x\ (1 + i\ x\ n) + M_{total}$

Fixed Cost $= 8{,}000\ x\ (1 + 0.12\ x\ 3) + 2{,}400$

Fixed Cost $= 8{,}000 \times 1.36 + 2{,}400$

Fixed Cost $= 10{,}880 + 2{,}400$

Fixed Cost $= \mathbf{\$13{,}280}$

Total revenue needed & markup

Step 5 – Total revenue needed from reagent sales

$\text{Revenue}_{needed} = \text{Fixed Cost} + \text{COGS}_{total}$

$\text{Revenue}_{needed} = \$\mathbf{13{,}280} + \$\mathbf{4{,}021.55}$

$\text{Revenue}_{needed} = \$\mathbf{17{,}301.55}$

Step 6 – Markup factor (same percentage for all reagents)

$$\text{Markup} = \frac{\text{Revenue}_{needed}}{\text{COGS}_{total}}$$

Markup $= \$\mathbf{17{,}301.55}$ / 4,021.55

Markup = **4.3022**

New selling price per pack for reagent *j*

$$\boxed{P_{\text{sell},j} = P_{\text{cogs},j} \times \text{Markup}}$$

Multiply each COGS by **4.3022**:

As seen in the excel sheet 2

These prices ensure that over 3 years you recover:

- Analyzer cost ($8,000)
- 12% annual interest on locked capital ($2,880 total)
- Maintenance & service ($2,400)
- All reagent COGS ($4,021.55)
- **Zero additional profit** (break-even). If you want profit, multiply final prices by (1 + profit margin).

Insight: Low-volume labs pay much higher reagent prices – often commercially unviable. Recommend minimum monthly volume commitment.

Example 3 = SHEET 3

PBS Biochemistry Analyzer – Reagent Rental Pricing Formula

(Refer to the Excel Sheet Labelled 2)

Higher Analyzer Cost & Higher Maintenance

3-Year Placement |15 Parameters | Fixed Monthly Test Volumes

Numerical Calculation (with assumed $i = 12\%$, $M_{total} = \$2,400$)

Must Known Determing Factors

Parameters	**Symbol**	**Value**
Analyzer landed cost	C_a	\$12,000 (Variables) (more advanced biochemistry analyzer)
Contract period	n	3 years (Variables)
Annual cost of capital (interest rate)	i	12 % (assumed) * (Variables)
Total maintenance & service cost over 3 years	M_{total}	\$5,400 (assumed 15% of C_a per year) *
Operating months	$12n$	36 months (Variables)
Tests per month per parameter (from customer data)	T_{month}	102 (same for all 15 parameters)

3. **Step-by-Step Formula**

Step 1 – Total tests per reagent over 3 years

$$\text{Tests}_{total} = T_{month}\ x\ 12\ x\ n$$
$$= 102\ x\ 12\ x\ 3$$
$$= 3672 \text{ tests}$$

Step 2 – Number of packs needed for each reagent (rounded up)

$$Q_j = \left\lceil \frac{\text{Tests}_{total}}{R_j} \right\rceil$$

where R_j= tests per pack for reagent j (from table).

Total packs over 3 years: $\sum Q_j = 118$

Step 3 – Total reagent COGS over 3 years

Sum of $Q_j \times P_{\text{cogs},j}$ (using given pack prices as COGS):

Total reagent COGS over 3 years

$$\text{COGS}_{total} = \sum_{j=1}^{15} (Q_j \times P_{\text{cogs},j})$$

where $P_{\text{cogs},j}$= your cost per pack (given in table).

COGS_{total}= **\$7970.59**

Fixed cost to recover

Step 4 – Total fixed cost to recover (analyzer + interest + maintenance)

Fixed Cost $= C_a \, x \, (1 + i \, x \, n) + M_{total}$

Fixed Cost $= 12{,}000 \, x \, (1 + 0.12 \, x \, 3) + 5{,}400$

Fixed Cost $= 12{,}000 \times 1.36 + 5{,}400$

Fixed Cost $= 16{,}320 + 5{,}400$

Fixed Cost $= \$\mathbf{21,720}$

Total revenue needed & markup

Step 5 – Total revenue needed from reagent sales

$\text{Revenue}_{needed} = \text{Fixed Cost} + \text{COGS}_{total}$

$\text{Revenue}_{needed} = \$\mathbf{21,720} + \$\mathbf{7970.59}$

$\text{Revenue}_{needed} = \$\mathbf{29,690.59}$

Step 6 – Markup factor (same percentage for all reagents)

$$\text{Markup} = \frac{\text{Revenue}_{needed}}{\text{COGS}_{total}}$$

Markup $= \$\mathbf{129,690.59} \,/\, \mathbf{7970.59}$

Markup = **3.725**

New selling price per pack for reagent *j*

$$\boxed{P_{\text{sell},j} = P_{\text{cogs},j} \times \text{Markup}}$$

Multiply each COGS by **3.725**:

As seen in the excel sheet 3

Example 4 = SHEET 4

Total revenue needed & markup

Step 5 – Total revenue needed from reagent sales

$\text{Revenue}_{needed} = \text{Fixed Cost} + \text{COGS}_{total}$

$\text{Revenue}_{needed} = \$13{,}280 + \$7{,}809.31$

Total cost $\text{Revenue}_{needed} = \$21{,}089.31$

Profit = $0.25 \times 21{,}089.31 = 5{,}272.33$

Total $\text{Revenue}_{needed} = 21{,}089.31 + 5{,}272.33 = 26{,}361.64$

$$\text{Markup} = \frac{\text{Revenue}_{needed}}{\text{COGS}_{total}}$$

Markup over COGS = $\frac{26{,}361.64}{7{,}809.31} = 3.376$

New selling price per pack for reagent *j*

$\boxed{P_{\text{sell},j} = P_{\text{cogs},j} \times \text{Markup}}$

Multiply each COGS by **3.725**:

As seen in the excel sheet 3

Example 4 = SHEET 4

PBS Biochemistry Analyzer – Reagent Rental Pricing Formula

Base Case + 25% Profit Margin on Total Cost
Now add a **25% profit margin** on the **total cost** (analyzer + interest + maintenance + reagent COGS).

(Refer to the Excel Sheet Labelled Labelled 1)

3-Year Placement |15 Parameters | Fixed Monthly Test Volumes

Numerical Calculation (with assumed $i = 12\%$, $M_{total} = \$2,400$)

Must Known Determing Factors

Parameters	**Symbol**	**Value**
Analyzer landed cost	C_a	\$8,000 (Variables)
Contract period	n	3 years (Variables)
Annual cost of capital (interest rate)	i	12% (assumed) * (Variables)
Total maintenance & service cost over 3 years	M_{total}	\$2,400 (assumed 10% of $C_a$per year) *

Operating months	$12n$	36 months (Variables)
Tests per month per parameter (from customer data)	T_{month}	102 (same for all 15 parameters)

1. **Step-by-Step Formula**

Step 1 – Total tests per reagent over 3 years

$$\begin{aligned}\text{Tests}_{total} &= T_{month}\ x\ 12\ x\ n \\ &= 102\ x\ 12\ x\ 3 \\ &= 3{,}672 \text{ tests}\end{aligned}$$

Step 2 – Number of packs needed for each reagent (rounded up)

$$Q_j = \left\lceil \frac{\text{Tests}_{total}}{R_j} \right\rceil$$

where R_j= tests per pack for reagent j (from table).

Total packs over 3 years: $\sum \boldsymbol{Q_j} = \mathbf{114}$

Step 3 – Total reagent COGS over 3 years

Sum of $Q_j \times P_{\text{cogs},j}$ (using given pack prices as COGS):

Total reagent COGS over 3 years

$$\text{COGS}_{total} = \sum_{j=1}^{15} \left(Q_j \times P_{\text{cogs},j}\right)$$

where $P_{\text{cogs},j}$= your cost per pack (given in table).

COGS_{total}= **$7,809.31**

Fixed cost to recover

Step 4 – Total fixed cost to recover (analyzer + interest + maintenance)

$$\text{Fixed Cost} = C_a\ x\ (1 + i\ x\ n) + M_{total}$$

$$\text{Fixed Cost} = 8{,}000\ x\ (1 + 0.12\ x\ 3) + 2{,}400$$

$$\text{Fixed Cost} = 8{,}000 \times 1.36 + 2{,}400$$

$$\text{Fixed Cost} = 10{,}880 + 2{,}400$$

$$\textbf{Fixed Cost} = \$\mathbf{13,280}$$

Total revenue needed & markup

Step 5 – Total revenue needed from reagent sales

$$\text{Revenue}_{needed} = \text{Fixed Cost} + \text{COGS}_{total}$$

$$\text{Revenue}_{needed} = \$\mathbf{13,280} + \$\mathbf{7,809.31}$$

Revenue$_{needed}$ $= \$\mathbf{21,089.13}$

Total cost Revenue$_{needed}$ $= \$21{,}089.31$

Profit = $0.25 \times 21{,}089.31 = 5{,}272.33$

Total Revenue$_{needed}$$= 21{,}089.31 + 5{,}272.33 = 26{,}361.64$

$$\text{Markup} = \frac{\text{Revenue}_{needed}}{\text{COGS}_{total}}$$

Markup over COGS = $\frac{26{,}361.64}{7{,}809.31} = 3.376$

New selling price per pack for reagent *j*

$$\boxed{P_{\text{sell},j} = P_{\text{cogs},j} \times \text{Markup}}$$